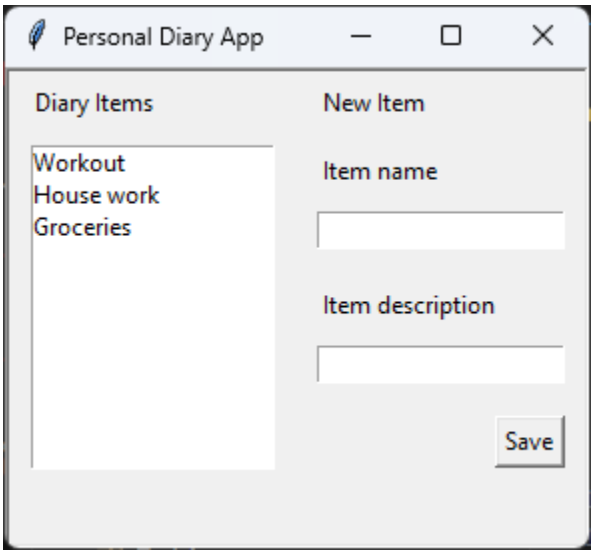
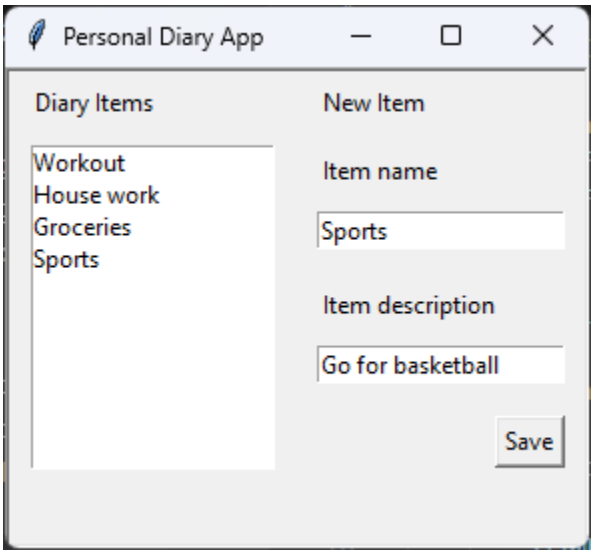
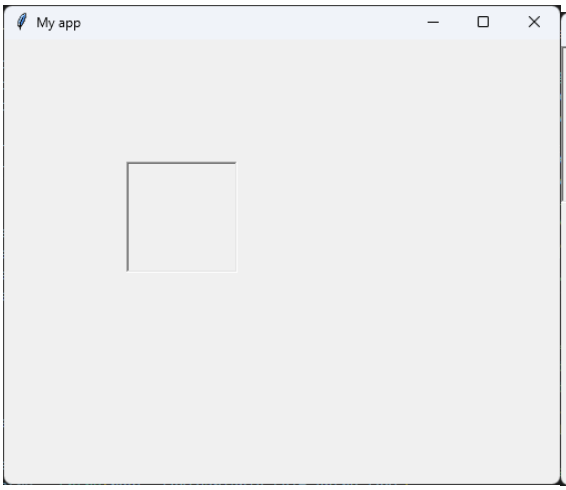
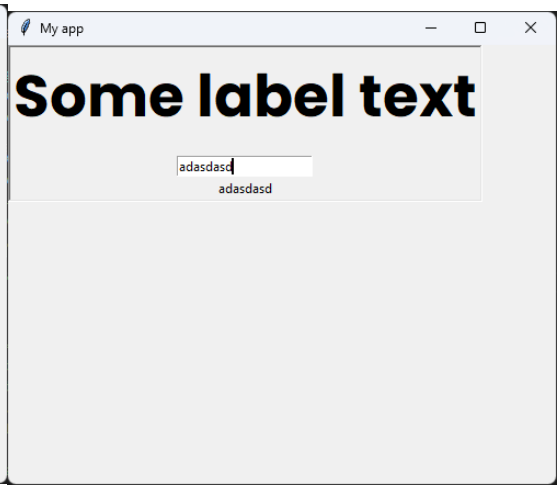
Task 1: Introduction to Tkinter







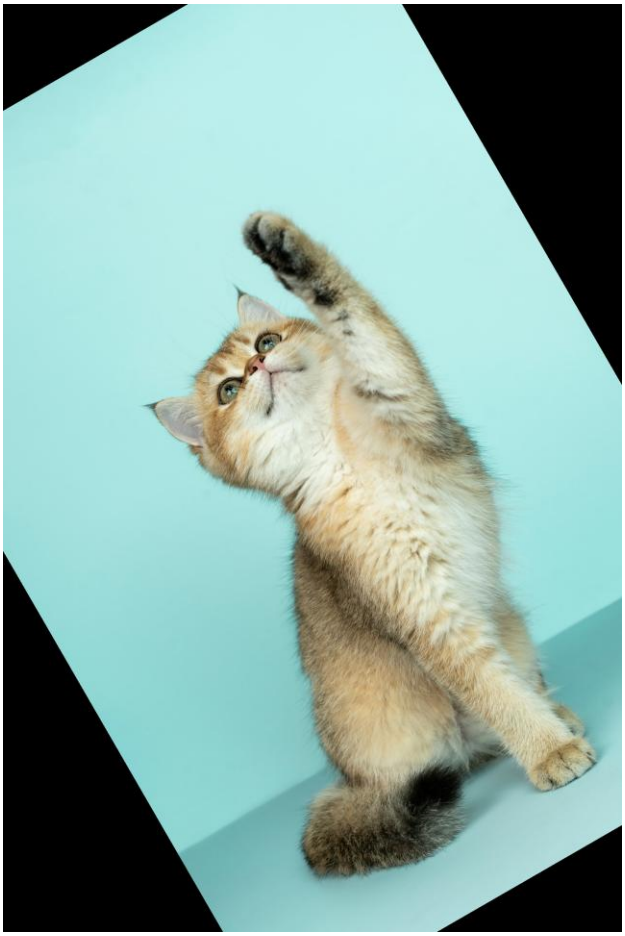


Task 2: Introduction to Pillow library/package

intro.py

intro.py > ...

```python
from PIL import Image

image = Image.open('Assets/cat.jpg')
# image.show()

# print(image.size)
# print(image.filename)
# print(image.format)

# hflipped_img = image.transpose(Image.Transpose.FLIP_LEFT_RIGHT).show()
# vflipped_img = image.transpose(Image.Transpose.FLIP_TOP_BOTTOM).show()
# rotated_img = image.transpose(Image.Transpose.ROTATE_90).show()

image.rotate(30).save("Output/rotatedCat.png", "PNG")
```

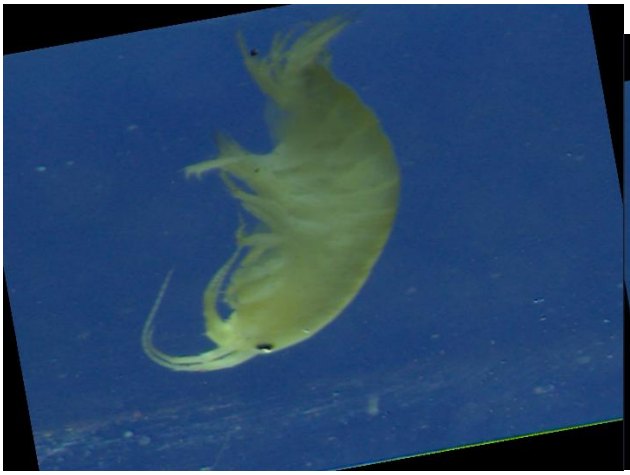


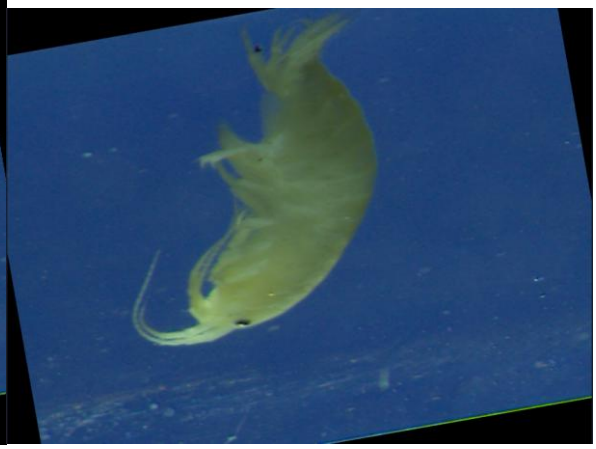
Task 3: Manipulating Images with Pillow/PIL library functions

intro.py | manipulate.py

manipulate.py > ...

```python
from PIL import Image
# import an image
image1 = Image.open('Assets/insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.PNG')
image2 = Image.open('Assets/insects/1_Gammar 2021_1_2021_06_03-13-19-23-898.PNG')
image3 = Image.open('Assets/insects/1_Limniu 2021_1_2021_06_02-13-08-37-958.PNG')

# flip
image_transpose = image1.transpose(Image.Transpose.ROTATE_90)
image_transpose = image2.transpose(Image.Transpose.ROTATE_90)
image_transpose = image3.transpose(Image.Transpose.ROTATE_90)
# rotation
image_rotate = image1.rotate(45, expand = False, center = (0,0))
image_rotate = image2.rotate(45, expand = False, center = (0,0))
image_rotate = image3.rotate(45, expand = False, center = (0,0))
# crop
image_crop = image1.crop((800,600,1600,1600))
image_crop = image2.crop((800,600,1600,1600))
image_crop = image3.crop((800,600,1600,1600))
# resize
image_resize = image1.resize((1000,600))
image_resize = image2.resize((1000,600))
image_resize = image3.resize((1000,600))
# applying multiple transformations to an image in a single line of
#code
combined_image1 = image1.transpose(Image.Transpose.ROTATE_90).resize((2000,1500)).rotate(10)
combined_image1.show()
combined_image2 = image2.transpose(Image.Transpose.ROTATE_90).resize((2000,1500)).rotate(10)
combined_image2.show()
combined_image3 = image3.transpose(Image.Transpose.ROTATE_90).resize((2000,1500)).rotate(10)
combined_image3.show()
```





Task 4: Understanding Pixels and Colors with Pillow/PIL library functions

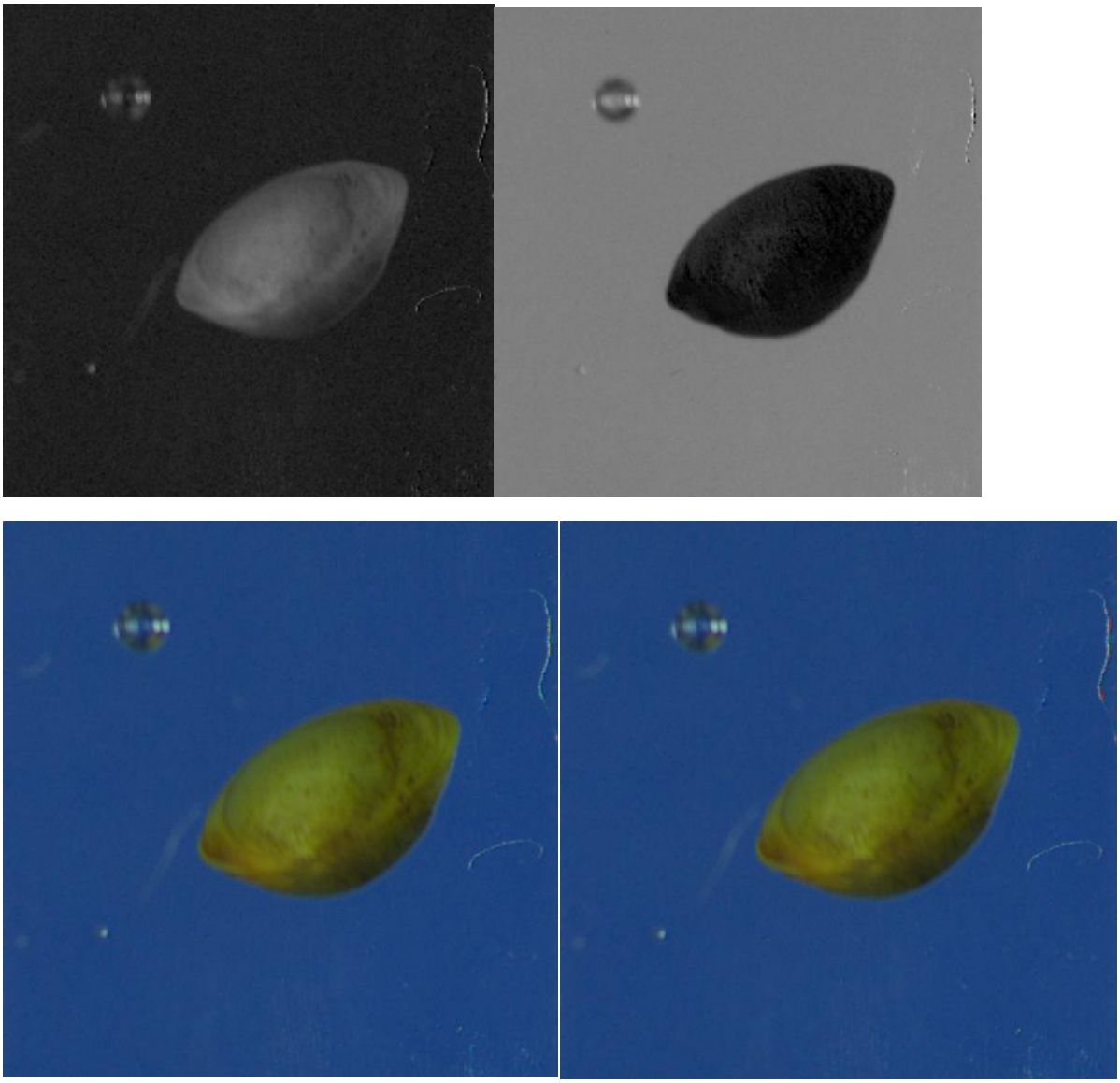
intro.py | manipulate.py | explore_colors.py | enhance.py

explore_colors.py > ...

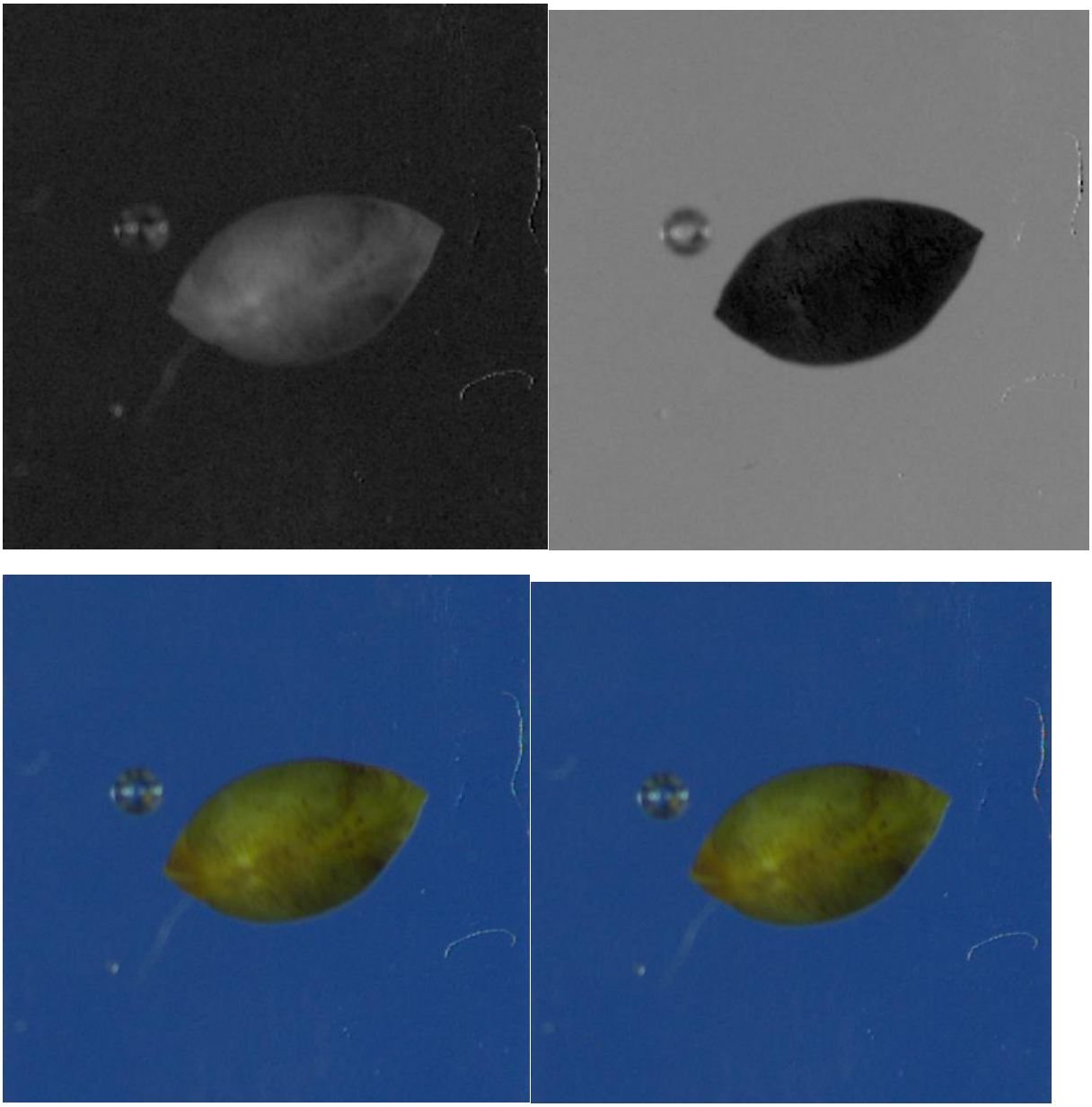
```python
from PIL import Image
image = Image.open('Assets/Insects/1_Limniu 2021_1_2021_06_02-13-08-37-981.PNG')
# get one pixel
print(image.getpixel((0,0)))

# # grayscale images
red_grayscale_image = image.getchannel('R')
blue_grayscale_image = image.getchannel('B')

# red_grayscale_image.show()
# blue_grayscale_image.show()

# change the pixel color
image.putpixel((0,0), (255,0,255))
# image.show()

# explore
print(image.getpixel((0,0)))
for x in range(image.size[0]):
    for y in range(image.size[1]):
        if image.getpixel((x,y))[0] == 0:
            image.putpixel((x,y), (200,20,20))
image.show()
```

2_Sphaer 2021_1_2021_06_03-11-22-54-894.PNG

```
$ "C:/Program Files/Python313/python.exe" "c:/Users/spyro/Documents/OneDrive - University of Canberra - STUDENT/Personal/SoftwareT1/Week4 Lab/explore_colors.py"
(36, 73, 124)
(255, 0, 255)
```



2_Sphaer 2021_1_2021_06_03-11-22-54-873.PNG



1_Limniu 2021_1_2021_06_02-13-08-37-981.PNG

Task 5: Image Enhancement with Pillow/PIL library functions

intro.py | manipulate.py | explore_colors.py | enhance.py

enhance.py > ...

```python
from PIL import Image, ImageEnhance
# import an image
image1 = Image.open('Assets/insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.PNG')
image2 = Image.open('Assets/insects/1_Gammar 2021_1_2021_06_03-13-19-23-898.PNG')
image3 = Image.open('Assets/insects/1_Limniu 2021_1_2021_06_02-13-08-37-958.PNG')
# create an enhancer
# vibrance_enhancer = ImageEnhance.Color(image)
contrast_enhancer = ImageEnhance.Contrast(image1)
brightness_enhancer = ImageEnhance.Brightness(image2)
sharpness_enhancer = ImageEnhance.Sharpness(image3)
# apply the enhancer
enhanced_image1 = contrast_enhancer.enhance(1.5)
enhanced_image2 = brightness_enhancer.enhance(2)
enhanced_image3 = sharpness_enhancer.enhance(3)
# show
image1.show()
enhanced_image1.show()
image2.show()
enhanced_image2.show()
image3.show()
enhanced_image3.show()
```

1_Gammar 2021_1_2021_06_03-13-19-23-879.PNG

1_Gammar 2021_1_2021_06_03-13-19-23-898.PNG

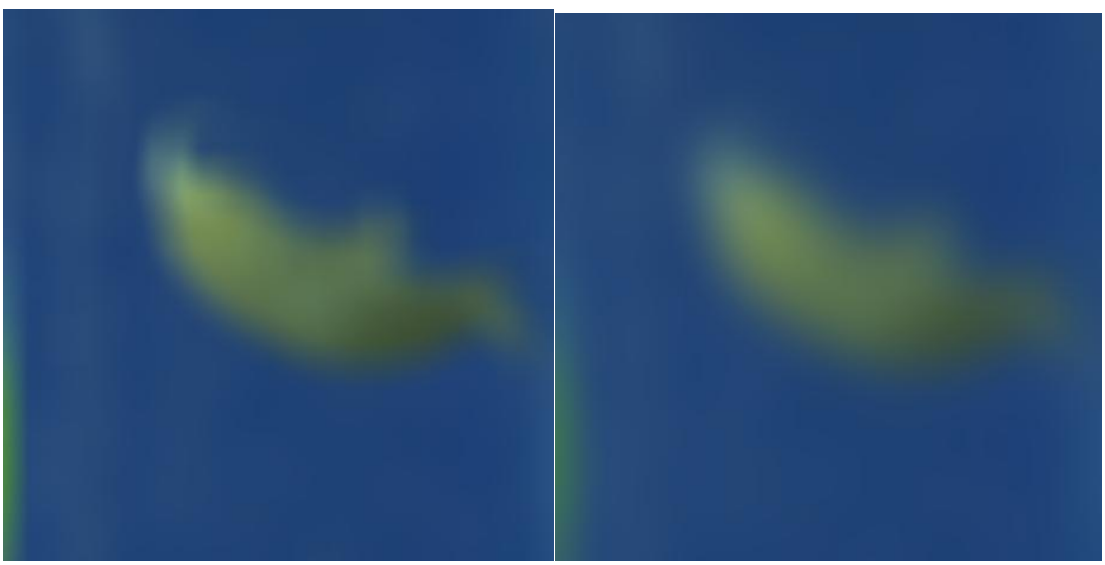
1_Limniu 2021_1_2021_06_02-13-08-37-958.PNG

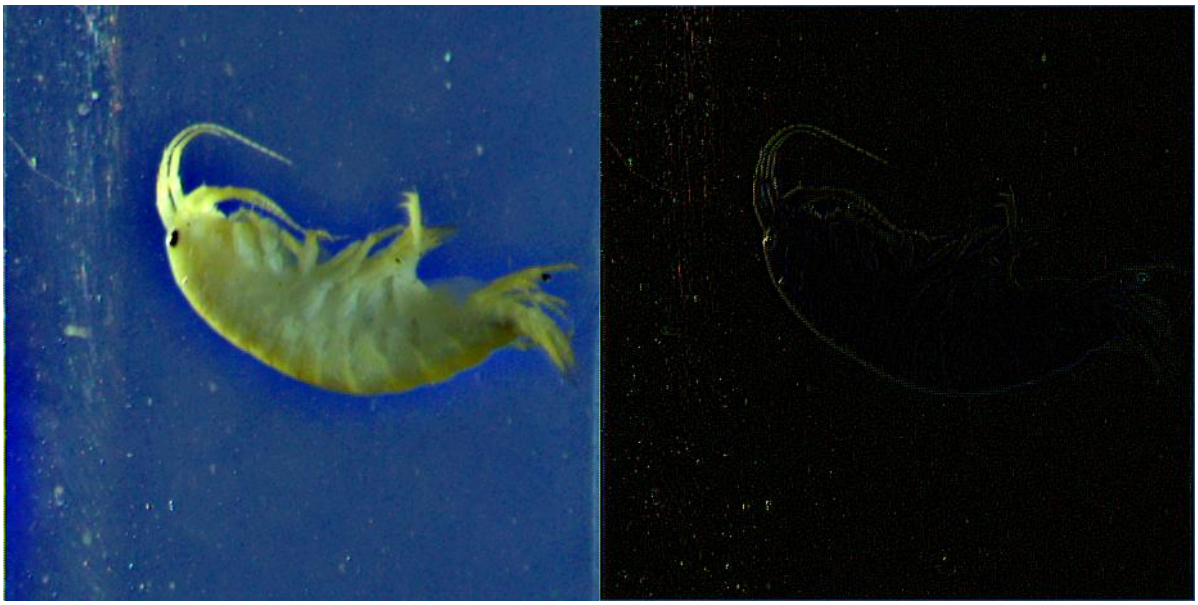
Task 6: Image filtering and Edge detection with Pillow/PIL library functions

intro.py | manipulate.py | explore_colors.py | enhance.py | image_filter.py

image_filter.py > ...

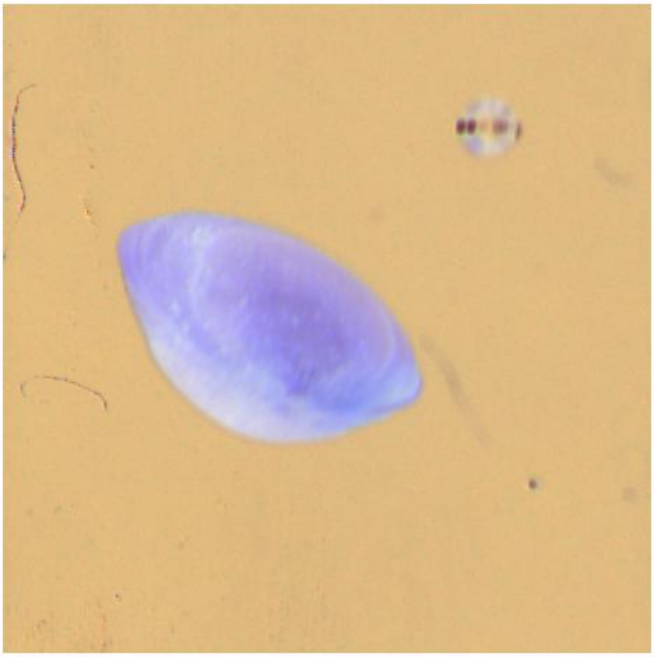
```python
from PIL import Image, ImageFilter
# create an image
image = Image.open('Assets/insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.PNG')
# apply a basic filter
# image_blur = image.filter(ImageFilter.BLUR)
# image_contour = image.filter(ImageFilter.CONTOUR)
# image_emboss = image.filter(ImageFilter.EMBOSS)
# image_edge = image.filter(ImageFilter.FIND_EDGES)
# apply advanced filters
image_boxblur = image.filter(ImageFilter.BoxBlur(radius = 20))
image_gaussianblur = image.filter(ImageFilter.GaussianBlur(radius
= 20))
image_unsharp = image.filter(ImageFilter.UnsharpMask(radius = 20))
image_boxblur.show()
image_gaussianblur.show()
```





Task 7: Image Operations with Pillow/PIL library functions

intro.py | manipulate.py | explore_colors.py | enhance.py | image_filter.py | image_operations.py

image_operations.py > ...

```python
from PIL import Image, ImageOps
# create an image
image = Image.open('Assets/insects/1_Limniu 2021_1_2021_06_02-13-08-37-958.PNG')
# position changes
image_mirror = ImageOps.mirror(image)
# image_scale = ImageOps.scale(image, 0.5)
# color changes
image_inverted = ImageOps.invert(image_mirror)
# add and remove
image_border = ImageOps.expand( image = image_inverted,  border = 50, fill = (255,255,255))
# image_padded = ImageOps.pad(image, (4000,6000))
# image_crop = ImageOps.crop(image = image, border = 200)
image_border.show()
```